REGIONAL ADIMINISTRATION AND LOCAL GOVERNMENT

LONGIDO DISTRICT COUNCIL

FORM THREE

MARKING GUIDE -MARCH 2024

036 INFORMATION AND COMPUTER STUDIES

1. **@1marks**

I.	ii..	iii.	iv.	v.	vi.	vii.	viii.	ix.	x.
B	C	B	D	A	A	C	B	C	A

2. **@1marks**

LISTA	i	ii	iii	iv	v
LIST B	C	H	G	D	B

3. (a) Printing is the process of producing text or images on paper or other materials using a printer. (**2marks)**

(b) Two possible causes for the printer failing to give output print could be: (**4marks)**

(i) Connectivity issues between the desktop computer and the HP LaserJet printer , such as a loose cable connection or network configuration problems.

(ii) Printer driver issues, such as outdated or corrupted printer drivers on the desktop computer, which can prevent the printer from functioning correctly .

(c) Two types of printers are: (**4marks)**

(i) Inkjet printers: These printers use liquid ink sprayed through microscopic nozzles onto the paper to create images.

(ii)Laser printers: These printers use a laser beam to produce an image on a drum, which is then transferred onto paper using toner and heat.

4(a) URL stands for Uniform Resource Locator. It is a web address that specifies the location of a resource on the internet. (**1marks)**

(b) Parts of the URL: (**4 marks)**

http:// - Protocol or scheme

www - Subdomain (optional)

cardealersTZ - Domain name

com - Top-level domain (TLD)

(c) An Internet Service Provider (ISP) is a company that provides individuals and organizations with access to the internet and related services. **(1marks)**

(d) Four examples of Internet Service Providers (ISPs) companies found in Tanzania could be: **(4marks)**

- Vodacom Tanzania
- Airtel Tanzania
- Tigo Tanzania
- Zantel

5. Three physical network layouts include: drawing is required **(6marks)**

(a) (i) Bus Topology: In this layout, all devices are connected to a single backbone cable, forming a linear structure. Data is transmitted through the cable, and each device receives the data but only processes information intended for it.

(ii) Star Topology: In this layout, all devices are connected to a central hub or switch, forming a star-like structure. Data passes through the hub, and communication between devices occurs through the hub.

(iii) Ring Topology: In this layout, each device is connected to two other devices, forming a circular structure. Data travels in one direction around the ring until it reaches its destination.

(b) Two types of networks based on area coverage: **(4marks)**

(i) LAN (Local Area Network): A LAN covers a small geographical area, typically within a single building or campus. It allows for the sharing of resources like files, printers, and internet connections among connected devices.

(ii) WAN (Wide Area Network): A WAN spans a large geographical area, often across cities, countries, or even continents. It connects multiple LANs and other networks, enabling

communication over long distances, typically using leased lines or public networks like the internet.

6 (a) Signs of a computer affected by a virus include: (**4marks)**

(i) Slow Performance: The computer may run noticeably slower than usual due to the virus consuming system resources.
(ii) Frequent Crashes: Unexpected system crashes or freezes may occur as the virus interferes with normal system operations.
(iii) Unusual Pop-ups: The appearance of unexpected pop-up windows or advertisements could indicate a virus infection.
(iv) File Corruption: Files may become corrupted or inaccessible, leading to data loss or errors when trying to open them.

(b) Ways for spreading computer viruses include: (**4marks)**

(i) Email Attachments: Viruses can be spread through malicious email attachments, often disguised as legitimate files or documents.
(ii) Infected Websites: Visiting compromised websites or downloading files from untrusted sources can lead to virus infections.
(iii) Removable Media: Viruses can spread through USB drives, CDs, or other removable media devices if they contain infected files.
(iv) Social Engineering: Cybercriminals may use social engineering tactics, such as phishing emails or fake software downloads, to trick users into installing viruses unknowingly.

(c) Two popular antivirus programs are: (**2marks)**

(i) Norton Antivirus: Developed by NortonLifeLock, Norton Antivirus provides real-time protection against malware, viruses, and other online threats.
(ii) McAfee Antivirus: McAfee Antivirus offers comprehensive protection against viruses, spyware, ransomware, and other malicious software, with features like firewall protection and web browsing security.

7 (a) Power disturbances over a computer include: (**6marks)**

(i) Surge: A surge is a sudden increase in voltage above the normal level, often caused by lightning strikes, power grid fluctuations, or switching on high-powered electrical devices.

(ii) Sags: Sags, also known as dips or brownouts, are temporary decreases in voltage below the normal level, which can occur due to power grid issues, overloaded circuits, or sudden power demands.

(iii) Outage: An outage is a complete loss of power, resulting from events such as power grid failures, equipment malfunctions, or intentional shutdowns.

(b) Two computer power management devices are(**2marks)**:

(i) Uninterruptible Power Supply (UPS): A UPS provides backup power to connected devices in the event of a power outage or fluctuation. It typically includes a battery that kicks in automatically to keep devices running until normal power is restored.

(ii) Voltage Regulator: A voltage regulator stabilizes the voltage supplied to computer equipment, ensuring consistent and safe levels of electrical power. It helps protect devices from damage caused by voltage fluctuations.

(c) Effects of voltage fluctuations to electrical appliances include: (**2marks)**

(i) Damage: Voltage fluctuations can cause damage to electrical appliances by subjecting them to higher or lower voltages than they are designed to handle, leading to component failure or reduced lifespan.

(ii) Data Loss: Sudden power disruptions or fluctuations can result in data loss or corruption on electronic devices such as computers or storage drives if data is not properly saved or protected.

8

(i) =SUM(E6:K6) (**2marks)**

(ii) 74(**2marks)**

(iii)=AVERAGE(E8:K8) (**2marks)**

(iv)E10(**2marks)**

(a) Column (**1marks)**

(b) Row (**1marks)**

9 (a) **Data backup** refers to the process of copying and storing important data to prevent loss in case of hardware failure, accidental deletion, or other unforeseen events. (**2marks)**

(b) Three importance of data backup are: (**3marks)**

(i) Disaster recovery: Data backup ensures that important information can be restored quickly and efficiently in case of hardware failure, natural disasters, or cyber attacks.

(ii) Data preservation: Backup helps in preserving valuable data, including documents, photos, videos, and other digital assets, reducing the risk of permanent loss.

(iii) Business continuity: For businesses, data backup is crucial for maintaining operations and minimizing downtime, enabling them to continue serving customers even in the face of disruptions.

(c) Five computer devices used for data backup are: (**5marks)**

(i) External hard drives
(ii) USB flash drives
(iii) Network-attached storage (NAS) devices
(iv) Tape drives
(v) Cloud storage services

10Advantages of E-commerce in Tanzanian Society**:(introduction 3, point with example 10 conclution 02)**

points

(i) Increased Access to Goods and Services
(ii) Convenience.
(iii) Job Creation
(iv) Empowerment of Small Businesses.
(v) Financial Inclusion

11Role of ICT in Various Sectors: **:(introduction 3, point with example 10 conclution 02)**

(i) Agriculture:
(ii) Medicine:
(iii) Entertainment
(iv) Engineering:
(v) Education